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Ruth Quinn
Alejandro Dussaillant J.

Civil Engineering Department, University of Greenwich, Chatham, Kent, UK

Research Article

Modeling Heavy Metal Behavior in Sustainable Drainage Systems: A Case Study

A major problem of increased urbanization is the rise in pollution caused by run-off. Among alternative management strategies, the use of sustainable drainage systems (SuDS) such as rain gardens and other bioretention facilities is becoming an increasingly attractive option to reduce these problems. However, there are few tools if any available for their design considering pollutant retention. In this paper, a dual-permeability based model that predicts the fate of heavy metals in SuDS is presented, and applied to the design of a rain garden system for a planned roundabout in Kent, UK. Preliminary design considered an upper root zone layer with organic soil and a sandy storage sub-layer, each 30 cm thick, for a bioretention area of 5 and 10% the size of the contributing impervious surface. Two scenarios are examined: the accumulation and movement of metals without macropores and the possibility of groundwater contamination due to preferential flow. It is shown that levels of lead can build up in the upper layers of the system, but only constitute a health hazard (surpass UK standard of 750 mg/kg) after 10 years. Simulations show that copper was successfully retained (no significant concentrations below 50 cm of rain garden soil depth). Finally, given concerns of preferential flow bypassing bioretention facilities, macropore flow was examined. Results indicated that due to site conditions it was not a threat to groundwater in the timeframe considered.

Keywords: Bioretention; BMP; Dual-permeability model; Rain garden; SUDS

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1 Introduction

In recent years, focus has been placed on alleviating the consequences caused by urbanization such as urban flooding, groundwater depletion and increased diffuse pollution. This has resulted in the emergence of sustainable drainage systems (SuDS) such as rain gardens, as a method of alleviating the problems mentioned above. A rain garden is a vegetated depression that has been specifically designed to collect and infiltrate the stormwater runoff from impervious areas such as car parks, roofs, and pavements. They are usually shallow depressions (<20 cm in depth) and typically 5–20% the size of the impervious surface from which they receive runoff [1]. Their general design consists of a layered soil profile comprising of an organic matter rich root zone and underlying storage/transmission zone containing sand and sometimes also gravel. They have numerous advantages such as increased groundwater recharge [2], pollutant retention [3], and have been shown to reduce peak storm flow volume [4].

Although emphasis is placed on decreasing urban flooding in the proposed UK National Standards for sustainable drainage [5], it should be noted that reduced recharge is a significant problem where new urbanizations have increased imperviousness, often compounded by increased groundwater abstractions, such as the case in several countries including the USA. Additionally, increased recharge caused by leaking pipes is being diminished by increases in pipe reliability and pressures on water companies to reduce leaks [6]. In addition, although recharge is not the overall objective of these facilities in the UK, it is a beneficial consequence particularly in drier areas such as the South East, and is an imperative function of these systems in other countries. Rain gardens differ from infiltration basins as water will infiltrate within hours and they are planted with herbs, grasses and shrubs. They are not as prone to the problems associated with infiltration basins such as clogging by sediments and sealing of the basin floor given bioturbation of soil due to plant roots and soil macrofauna [7]. The evapotranspiration from the vegetation in a rain garden also provides a higher available soil water capacity for the next rainfall event allowing for increased infiltration and water storage in the soil profile. Currently, they are not as prevalent in the UK as in other countries such as the USA. However, their popularity in England and Wales is certain to increase with the introduction of legislation such as the Flood and Water Management Act [8], which promotes the use of SuDS in new developments and redevelopments by requiring drainage systems to be approved against specified proposed National Standards [5].

Numerical computer models have been developed to quantify and optimize the extent of groundwater recharge by these facilities; models RECHARGE [2], RECARGA [1], and R2D [9]. These models have been used both to design rain gardens and also for research purposes

Correspondence: Ruth Quinn, P061, Pembroke Building, University of Greenwich, Central Avenue, Chatham Maritime, Kent, ME4 4TB, UK
E-mail: r.quinn@greenwich.ac.uk

Abbreviations: **KWE**, kinematic wave equation; **SuDS**, sustainable drainage systems where they were applied to examine the effects of different design parameters on recharge thus enabling optimization of these facilities [1, 10]. However, these models do not simulate pollutant loading and removal, which is one of the main reasons why the model discussed in this paper was developed. To the authors' knowledge, no other models exist that can sucessfully and expeditiously predict accumulation and transfer of metals through these facilities over long periods of time. There is a need for such a model as groundwater contamination is a serious concern for design engineers especially in areas of shallow groundwater or protection zones.

Heavy metals were chosen as the initial focus of this model as they pose the greatest health hazard to humans and animals owing to their accumulation in the upper layers of these facilities. The principal heavy metals found in stormwater runoff are copper (Cu), lead (Pb), and zinc (Zn). The primary source of these metals is vehicle deterioration such as the wear of brake pads and tyres. Other sources include metal coated (Zn and Cu) roofs along with industrial runoff [11]. The behavior of these metals in bioretention facilities has been thoroughly documented [3, 12]. In the majority of cases, metals were removed almost completely [13] with the vast majority of this removal occurring in the first 25 cm of soil [14]. This indicates that heavy metals accumulate in the upper layers of the system. It is important to quantify this accumulation as the Pb concentrations in the upper layer can exceed regulatory guidelines for soil metal concentrations in residential areas [14]. This may require the removal and replacement of this contaminated soil, which can be a costly procedure and damaging to the established flora and fauna. Thus, it is an essential consideration at the design stage so as to optimize soil replacement intervals.

Another phenomenon, which must be considered regarding fate and transport of pollutants is macropore flow. Macropores are large continuous openings in soil; flow rates through these openings can be significantly greater than in the surrounding area when soil is near saturation. This can result in the rapid downward movement of water and solutes including pollutants through the system [15]. As heavy metal capture is mainly observed in the upper layers of soil, runoff flowing through the system via macropores could bypass the adsorptive top stratum and could result in groundwater contamination. The formation of macropores has been attributed to numerous factors including activities of soil animals and elongation or decay of plant roots. The effect of physical processes on this formation has been studied mainly using column experiments where it was also found that both freeze–thaw and wet–dry cycles can increase the occurrence of macropores in soils [16].

Macropores are commonly reported in agricultural and roadside soil among others but to the authors' knowledge no research has been completed into macropore formation specifically in rain gardens [16]. However, it is certain that bioretention systems will develop macropores during their lifecycle owing to bioturbation, plant root, and soil macrofauna such as earthworms. It might not be advisable to prevent macropore formation in rain gardens as macropores can increase infiltration and percolation (both of which are desired functions of bioretention systems). Yet, it is important to be aware of potential issues, or trade-offs such as between infiltration/recharge and groundwater pollution risk. One alternative is to have a model that can be used to explore potential designs that can reduce these risks and find a reasonable balance between objectives.

A model was developed to predict the fate of heavy metals in SuDS and consider the significant issues highlighted above. In this paper, this model is applied to the design of a proposed rain garden in Thanet, Kent in the U.K. Heavy metal transport needs to be modeled in this system given the underlying saturated chalk and thus risk of groundwater contamination.

Two specific scenarios will be examined:

(1) Heavy metal build-up in the upper soil layers of the system without macropore flow.
(2) The extent of groundwater pollution due to macropore flow.

By completing this investigation, the advantages of using this model as a method for the design and analysis of bioretention facilities will be demonstrated.

2 Materials and methods

This model comprises of two parts; a heavy metal retention model and a hydraulic model both will be discussed below (more details available in [17]).

2.1 Heavy metal retention modeling

Adsorption was identified as the dominant mechanism by which heavy metals are removed from runoff [13]. The other mechanism for removal, i.e. accumulation in vegetation is virtually negligible and found to account for only between 0.5 and 3.3% of the retention [13] and thus is not examined here. The model has incorporated several isotherms options for modeling pollutant adsorption to soil particles: the linear, Langmuir and Freundlich equations.

The linear isotherm assumes that the media-sorbed pollutant concentration (C_s) is directly proportional to the dissolved pollutant concentration (C) [14]:

$$C_s = K_d C \tag{1}$$

where K_d is the linear distribution coefficient.

The other most prevalent isotherms for heavy metal retention modeling are the Langmuir and Freundlich isotherms. The Langmuir isotherm assumes monolayer adsorption at higher input concentrations but at lower levels it essentially reduces to a linear isotherm. It has been proven accurate and used for a wide range of adsorption cases however it does have a number of disadvantages. It does not account for the surface roughness of particles in the adsorbate [18]. In some facilities, coarse sand is used in the upper layers thus the Langmuir isotherm may not always be appropriate [19]. The Freundlich isotherm however models multilayer adsorption and can predict non ideal sorption [18]. This equation has been criticized because it lacks a fundamental thermodynamic basis as it does not reduce to Henry's law at low concentrations [18]. This could cause a problem as typically heavy metal concentrations in stormwater are low. Additionally, it is more complex than the other isotherms owing to the additional parameters needed.

Despite its simplicity, the linear isotherm has proven accurate for long term modeling over simulated periods in excess of 100 years [20]. In the experiments performed by the Highways Agency, 32 columns were exposed to infiltration regimes containing heavy metals and other pollutants that simulated 125–300 years of infiltration under typical climatic conditions. From these findings, K_d values were derived and the results compared with the linear, Langmuir and Freundlich isotherms. Although the Langmuir and

Freundlich isotherms provided the best fit to the data, the linear isotherm also provided satisfactory results ($R^2 > 0.78$).

In summary, given its simplicity and proven capabilities, it was decided to use the linear isotherm approximation for this study. However, the model can also work with one of the other isotherm approaches.

In order to combine adsorption with modeling the transport of heavy metals, the advection diffusion retardation equation is used:

$$R\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial z^2} - \upsilon\frac{\partial C}{\partial z} \tag{2}$$

where z and t are the positions in space and time, respectively, v is the pore water velocity, D is the diffusion coefficient and R is the retardation factor, which governs the retention of the pollutants and is dependent on the isotherm being used.

The value of R for the linear isotherm is as follows:

$$R = 1 + \frac{\rho}{\theta}K_{\mathrm{d}} \tag{3}$$

where ρ is the bulk density of soil and θ is the soil moisture content.

It should be noted that due to the heterogeneity of natural substances, the K_{d} value is unlikely to be constant throughout the layers of soil. However despite this, constant K_{d} values have been shown to provide accurate estimates for heavy metal retention [20].

Forms of the advection–diffusion–retardation have been used by various hydrological models (HYDRUS, MACRO) [21, 22]. However, it does have limitations, similar to the Richards and kinematic wave equations (KWE); it assumes no air entrapment that impedes water flow. Despite this, it has shown to be an accurate method of modeling heavy metal transport and retention [14].

2.2 Matrix and macropore modeling

A dual permeability approach is proposed for the modeling of water flow [21, 22] to account for both matrix and macropore flow.

In cases where gravity dominates vertical movement of soil moisture, such as drainage following infiltration and the water percolation deeper into soil, the KWE has been shown to suitably replicate water flux. The KWE is a reduction of the Richards equation whereby $\{D(\theta)\partial\theta/\partial z\}$ is assumed to be constant. Smith [23] accurately simulated unsaturated flow in layered heterogeneous soils similar to those that predominate in rain garden sub-soils using this method.

It is presented in one form as:

$$\frac{\partial h}{\partial t} + v\frac{\partial h}{\partial z} = 0 \tag{4}$$

where h is the soil–water head.

After examining several of the most common and well used existing dual permeability models (RZWQM, HYDRUS, MACRO), it was decided that the matrix and macropore regions should be modeled using this method.

By using a simple formula as opposed to the more complex Richards equation, it allows for long term modeling utilizing decades of hourly data, or several climate change scenarios in a shorter amount of computer processing time. It may experience slightly different results when compared to other hydrological models which utilize the Richards equation due to this simplicity; however this is more than compensated for by its utility [17]. It was found that the KWE when tested against the Richards equation had a coefficient of determination of 0.933 for the case of Pan and Wierenga [24]. However, the KWE takes approximately 45 min to run a ten year simulation using hourly data on a PC with an Intel Core i5 processer and 4.0 GB RAM whereas the Richards equation can take several days [7].

The soil moisture content is expressed using van Genuchten–Mualem equations [25].

As the KWE cannot calculate infiltration it is coupled with the Green and Ampt equation, which determines infiltration using the following formula [26]:

$$\frac{\mathrm{d}F(t)}{\mathrm{d}t} = i \quad 0 \le t \le t_{\mathrm{p}} \tag{5a}$$

$$\frac{\mathrm{d}F(t)}{\mathrm{d}t} = K_{\mathrm{sat}}\left(1 + \frac{B}{F(t)}\right) \quad t_{\mathrm{p}} \le t \le t_{\mathrm{r}} \tag{5b}$$

with:

$$B = (h_{\mathrm{wf}} + h_{\mathrm{s}}(t))(\theta_{\mathrm{sat}} - \theta_{\mathrm{ini}}) \tag{5c}$$

where F is the infiltration rate, i is the water supply intensity, K_{sat} is the saturated hydraulic conductivity, t_{p} is the time of water ponding, t_{r} is the time at which water input to the soil surface ends, h_{wf} is the average capillary suction head at the wetting front and $h_{\mathrm{s}}(t)$ is the ponded depth at the soil surface at time t. When t_{p} is reached macropore flow occurs in accordance with previous findings [27].

2.3 Validation

In order to facilitate the identification of any faults with the various components of the model, it was tested for each of its three parts: matrix, macropore, and pollutant retention for ease of validation.

The validation of the matrix region comprised of testing the model against two scenarios prominent in rain garden facilities. The first was a 24-h simulation of sharp wetting front carried out by Celia et al. [28] and the second was the layered soil profile case 1.2 of Pan and Wierenga [24]. Column experiments completed by Mdaghri and Germann [29] were chosen as suitable validation cases for the macropore segment of this model as the six runs which were performed on a layered soil encompass a range of infiltration, conductance and exponent value, the validation using run 3, is detailed in this paper. The linear isotherm method was validated against the experimental results of Davis et al. [3]. This dataset was chosen as the experiments were specifically focused on bioretention performance. They consisted on the heavy metal retention evaluation of several small columns (3.5 cm depth) filled with topsoil. Sample results from this validation are shown in Fig. 1. In order to determine the accuracy of the model several efficiency indexes were used to evaluate the goodness-of-fit: the coefficient of determination (R^2), the Nash–Sutcliffe (NS) coefficient and the root mean square error (RMSE), the results of which are shown in Tab. 1. Further details are available in [17].

2.4 Site description

The rain garden is to be constructed on a roundabout in Thanet, Kent in south eastern England and receive drainage from the road surrounding it. It is necessary to evaluate the potential pollutant

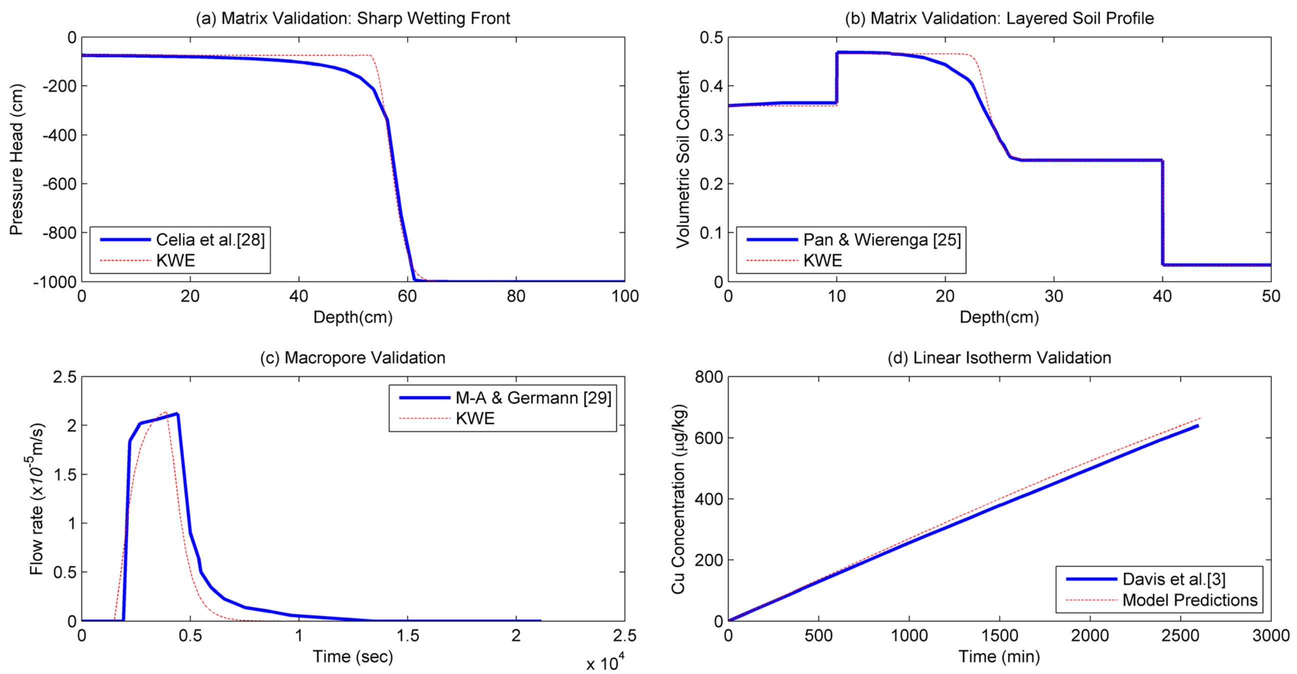

Figure 1. Validation of model with results from literature [3, 25, 28–29].

Table 1. Validation results

Subroutine	Reference	Coefficient of determination (R^2)	Nash–Sutcliffe (NS) efficiency index	Root square mean error (RSME)
Matrix region	[28]	0.996	0.995	30.5 cm (pressure head)
	[24]	0.931	0.921	0.038 m^3/m^3 (volumetric soil water content)
Macropore region	[29]	0.930	0.800	3.86×10^{-6} m/s (flow rate)
Pollutant retention	[3]	0.990	0.993	16 μg/kg (Cu soil concentration)

retention of the rain garden at this site as shallow groundwater is present. A model screening approach was viewed as the best method of achieving this and as an efficient and simple tool for informing discussion between Kent Council and the Environment Agency.

Several boreholes and detailed studies allow for a relatively good characterization of the subsoil, which is a required model parameter. The boreholes indicate an upper layer of made ground consisting of silty clay to 0.4 m and sandy gravel underlying to 1.8 m [30]. Groundwater was not encountered directly at the site however it was found that in the vicinity (100 m away) saturated chalk lay just 1.5 m deep [30]. The upper layer was heterogeneous as it consisted of several layers of made ground, however the lower layers of chalk are homogeneous. It is noted that there is a possibility of soluble rocks at the site as indicated by the British Geological Survey (www.bgs.ac.uk/products/geosure/soluble.html). It would thus be prudent to complete further site investigation to ensure the rain garden will not contribute to dissolution or degradation of the bedrock.

The specific soil parameters of the rain garden are obtained from Dussaillant [7] and are provided in Tab. 2. The depression depth is set at 15 cm in agreement with previous rain garden design [7].

The K_{sat} of the subsoil is 2 cm/h and was given by a soak-away test; further details are available in the site report [30]. Two distinct scenarios were modeled, firstly the transfer and accumulation of heavy metals without macropore flow and secondly, possible groundwater pollution caused by macropores.

A simulation period of 10 years was chosen as it was deemed to give an appropriate measure of time at which comprehensive maintenance may be expected. The hydrological data was given by a local weather station at Manston approximately three miles from the proposed site (http://badc.nerc.ac.uk/cgibin/midas_stations/station_details.cgi.py?id=775&db=midas_stations) and obtained from the British Atmospheric Data Centre (http://badc.nerc.ac.uk/view/badc.nerc.ac.uk__ATOM__dataent_ukmo-midas). The interval of 2003–2012 was chosen as it gave a sufficiently long period and included dry (2005) and wet (2009) years. The Pb input concentration

Table 2. Hydraulic parameters of the rain garden

Soil characteristic	Root zone	Storage zone layer
Texture	Loam	Sand
Depth (cm)	30	30
Saturated soil moisture content (θ_{sat}) (m^3/m^3)	0.41	0.41
Residual soil moisture content (θ_{res}) (m^3/m^3)	0.02	0.041
K_{sat} (cm/h)	10.16	15

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was specified as 0.08 mg/L and corresponds to the suggested level of synthetic runoff for rain garden design provided in Davis et al. [3]; it also matches Pb values found in runoff from a busy highway (120 000 vehicles per day) measured by Hares and Ward [31]. The Cu input concentration of 0.08 mg/L is also consistent with synthetic runoff values; this is double the observed value experienced at roadside sites [32]. Only Cu and Pb are examined in this study as they are most likely to pose contamination risk, Pb due to accumulation and Cu due to transfer through the system.

2.5 Accumulation and transfer without macropores

In this section, the accumulation of Pb in the proposed rain garden is evaluated for two upper layer soil substrates: an organically enriched soil (high retention capacity) commonly used in rain gardens and standard topsoil (low retention capacity) [3, 14]. This is reflected by the K_d values shown in Tab. 3. Area ratios are also considered in order to optimally design the facility and decrease the frequency of upper layer removal. Pb was chosen as the focus as it is the only metal known to exceed safety levels in rain garden soil [14]. Cu behavior is also examined as it is the most likely heavy metal to cause groundwater pollution given its high mobility in soil even with no macropores present and the fact that it has been found not to sorb as easily to soil as Pb or Zn [14]. As can be seen, the K_d values for Pb and Cu of standard topsoil are similar (500 and 550 L/kg, respectively). However for the organically enriched soil, the K_d value for Pb is significantly greater than Cu (171 214 L/kg in comparison with 4799 L/kg). This indicates that Pb will sorb to the soil at a much higher level than Cu.

The accumulation of Cu in soil does occur but does not pose a health hazard and thus is not examined in this paper [14].

2.6 Macropore flow

Typically macropores only occur in the root zone as this layer has the highest biological activity. The storage zone is designed to be coarse sand, without organic matter, to encourage flow and storage, which makes it relatively inert and discourages growth of roots, earthworm movement and other activities, which increase macropore flow. However, as groundwater contamination was the key concern, the worst case scenario of a macropore reaching 1.5 m (depth of the groundwater) is examined. This formation may take several years of development thus it is assumed that the macropore is active for only a seven year period. The macropore also receives the maximum inflow possible. The model uses the KWE to calculate macropore flow rate however the maximum flow rate in this case is limited to 2 cm/h owing to the maximum infiltration rate of the lower layer of chalk. This infiltration rate was measured using a soak-away test during the initial site investigation [30]. The flow supply is also limited by the depression depth. There is also a possibility of lateral flow from macropores to the matrix region; however as we have focused on conservative estimates geared to SuDS design, the worst case scenario is being simulated; this will not be examined here.

3 Results

Results for Pb concentration in the upper 10 cm of the rain garden with an area ratio (A) of 5 and 10%, respectively and highly retentive soil are shown in Fig. 2a. The maximum concentration in the rain

Table 3. Heavy metal retention parameters of the soil

	Linear distribution coefficient (K_d) (L/kg)				
	Pb		Cu		
Soil	High retention (organically-enriched)	Low retention (standard top soil)	High retention (organically-enriched)	Low retention (standard top soil)	Reference
Loam	171 214	500	4799	550	[3, 14]
Sand	1295		1060		[33, 34]

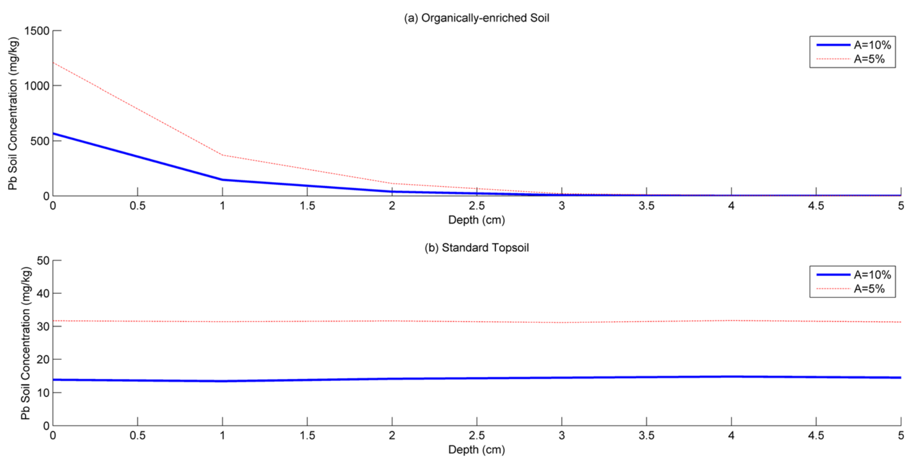

Figure 2. Lead concentration in rain garden soil with highly retentive (organically-enriched) soil ($K_d = 171\,214$ L/kg) and lower retentive (standard) top soil ($K_d = 500$ L/kg) and two area ratios.

garden with $A = 10\%$ is 568 mg/kg. However, if the area ratio is decreased to 5% to minimize construction costs, the level of lead accumulation increases significantly to 1209 mg/kg, higher than the allowable Pb concentration in soil of 750 mg/kg [30].

For soil with a lower K_d value, the accumulation in the upper layer of the systems for both area ratios is reduced as illustrated in Fig. 2b.

In addition to accumulation in the system, it is important also to consider the possibility of groundwater pollution caused by this lower retention. Figure 3 shows the simulation results for Pb concentration in soil–water for a lower Pb retention rate in the upper layer. It indicates that in the case of an area ratio of 5%, Pb does not reach a depth of <43 cm.

For the case of Cu, Fig. 4 illustrates results for concentration in soil-water through the soil profile for highly retentive soil with an area ratio of 5 and 10% after ten years. It shows that Cu soil–water concentration is negligible at 30 cm, which indicates that all retention is taking place in the upper layer. It also shows that area ratio has little effect on this concentration.

If however, the soil has a lower retention capacity such as normal topsoil (Tab. 3); it is observed that the majority of retention after ten years takes place in the lower storage layer (Fig. 5).

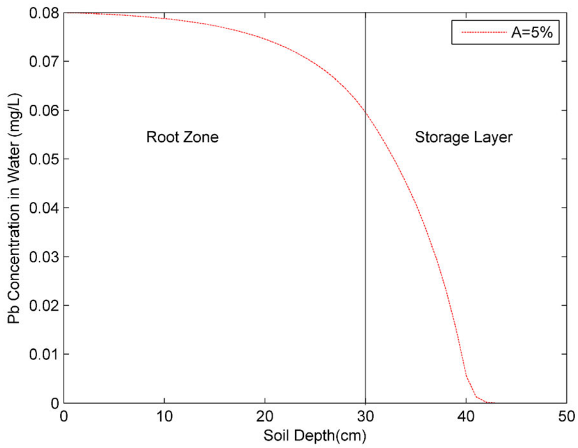

Figure 3. Lead concentration in water for lower retentive top soil with a 5% area ratio.

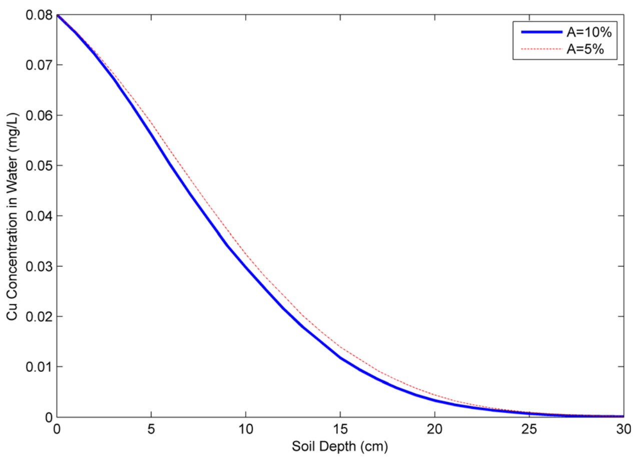

Figure 4. Copper concentration in soil–water for highly retentive (organically enriched) soil and two area ratios.

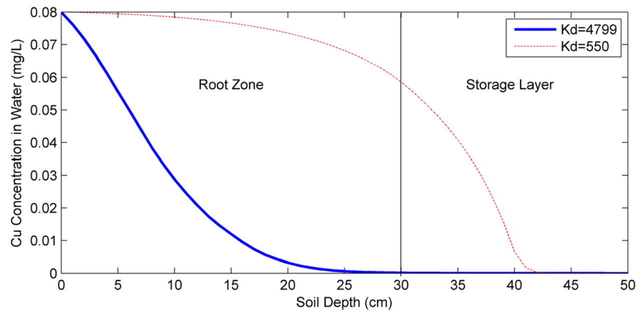

Figure 5. Copper concentration in water for high and lower retentive ($K_d = 550$ L/kg) soil with a 5% area ratio.

3.1 Macropore flow

In this section, the effect of macropore flow on groundwater pollution is examined. Figure 6 shows the Cu water concentration in a single macropore for soil with the higher retention value after a period of 10 years that has received the maximum water input based on the limiting factors stated above. While Fig. 7 shows the Cu water concentration when a soil with a lower quantity is used, here the no observable Cu is found after 55 cm for the worst case scenario (low retentive soil and an area ratio of 5%).

3.2 Sensitivity analysis

In order to quantify the effects of uncertainties in important parameter values, a sensitivity analysis was performed. The focuses of

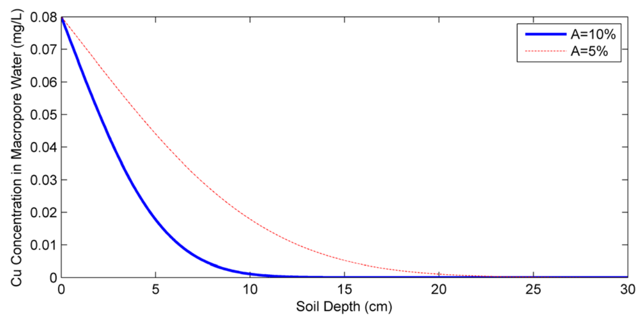

Figure 6. Copper concentration in macropore water for highly retentive (organically enriched) soil and two area ratios.

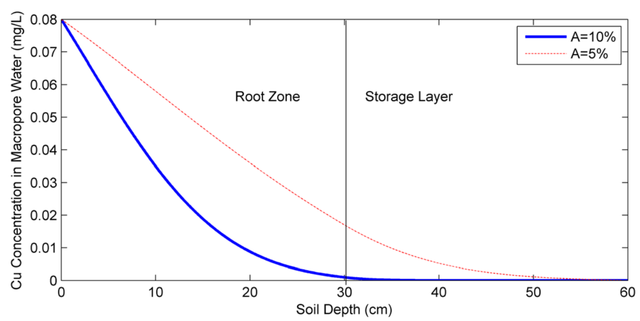

Figure 7. Copper water concentration in macropore water for lower retentive topsoil and two area ratios.

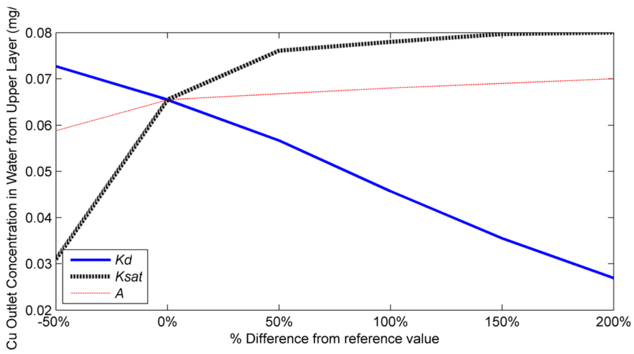

Figure 8. Sensitivity analysis of model to K_d, K_{sat}, and A.

this examination were the variables K_d, K_{sat}, and A as these have been shown to be the determining factor for retention. The reference values chosen for this analysis are those used earlier ($K_d = 550$ L/kg, $K_{sat} = 10.16$ cm/h, $A = 10\%$). As the main concern of this paper was to prevent groundwater contamination at the site, the effects of these variables on the Cu outlet concentration in water from the upper layer of the system were examined as demonstrated in Fig. 8.

A sensitivity analysis was not performed on macropore flow as the potential factors (number, velocity and capacity of macropores) would not significantly affect the results for outlet concentration given the maximum flow rate through the macropores of 2 cm/h determined by the lower layer of chalk. However, this could be performed for other situations.

4 Discussion

For highly retentive soils, the results for Pb above correspond well to experimental findings by Li and Davis [14] in that the majority of Pb retention is limited to the upper 10 cm of soil (Fig. 2a). The levels of accumulation are diminished when the value of K_d is reduced (Fig. 2b) and a more even distribution of lower Pb accumulation through the upper layer of the system is observed. Despite this, in all cases the facility demonstrated excellent performance in Pb removal and no groundwater contamination is predicted (Fig. 3). However, the risks posed by the accumulation of this metal require additional investigation. As the roundabout will not be accessible to the public, the soil pollutant guidelines for industrial and commercial sites provided by the Environmental Agency are appropriate; the regulatory level for Pb in these guidelines is 750 mg/kg [20]. This indicates that in the case of the lower area ratio (5%), the first 1 cm of soil needs to be replaced within the ten years period examined after approximately six years. This would cause a large degree of inconvenience as it requires vegetation be removed. The reason the contamination is limited to 1 cm is due to the high adsorption capacity of the soil, however this would not be the case in all circumstances. For example, if K_d is lower but still large enough to facilitate high rates of retention, a dangerous level of accumulation can be distributed to a greater depth. For $A = 10\%$, the level of accumulation does not exceed regulatory guidelines therefore no action needs to be taken.

If this rain garden was situated in a residential area, the maximum allowable concentration decreases to 350 mg/kg due to the possibility of human contact and thus would require remedial action at more regular intervals [14] even with an area ratio of 10%. This highlights the need to design rain gardens on an individual basis and further underlines the utility of this model.

With regards to Cu, again substantial retention is predicted, also in line with experimental evidence [29]. Results shown in Fig. 4 indicate that the water concentration values are minimally affected by a decreased area ratio. A possible reason for this is that a large increase in pore water velocity is required to have an effect on retention, an aspect, which needs to be further investigated. The majority of rainfall events at the site are of lower precipitation values ($<$10 mm) thus a significant rise in velocity is only seen during exceptional events minimally affecting retention.

When the K_d value of the upper layer is reduced, the majority of the capture shifts to the storage zone, but no metal flux is predicted from the rain garden to the subsoil and groundwater (Fig. 5). It is thus beneficial for the lower storage layer to also have high K_d values to ensure that despite metal flux from the upper layer no pollution reaches the subsoil.

In this case of macropore flow, no metal flux is predicted to enter the groundwater (Figs. 6 and 7). This is caused by the limiting infiltration rate of the chalk which decreases the maximum velocity in the macropore and thus increases sorption, sorption does not typically occur in the macropores as their water velocity can typically be in excess of 134 cm/s [26]. It is therefore important to consider each situation individually. Overall, these observations emphasize the relevance of this recently developed model, particularly given that macropores are not considered in the standards for SuDS design.

Regarding the sensitivity analysis (Fig. 8), a linear relationship between the outlet water concentration and the value of K_d exists: this is to be expected as the linear isotherm is used. Minor deviances from the reference value do not influence the results substantially, only when the value of K_d is altered by $>\pm 50\%$ are major effects ($\pm 10\%$) on the Cu outlet concentration seen. The outlet concentration is most sensitive to the value of saturated hydraulic conductivity K_{sat}. For low values of K_{sat}, outlet concentration is low, which may be due to increased overflow from the system and/or decreased pore water velocity through the soil. The analysis shows that varying the other parameters did not have a significant impact on results.

In summary, the model results imply that a higher distribution coefficient coupled with a low area ratio results in large levels of Pb accumulation. It thus may be necessary if interested in distributing the retention more evenly through the upper layers of the system, to use soils with lower Pb distribution coefficients. This would reduce the need for frequent costly removal of contaminated soil. Macropore flow seems not affect the groundwater quality given the limiting infiltration effect of the lower layer of chalk. Therefore, in this particular case, an area ratio of 5% is recommended to decrease construction and maintenance costs. A soil with a lower K_d value can be used for the upper layer to prevent hazardous accumulation, but care should be taken to ensure the storage layer has enough retention capacity itself to offset the decreased heavy metal retention this would cause.

It is clear from the findings of this paper that it is of vital importance to monitor the fate of heavy metals in SuDS during the design process, to ensure the creation of an optimum facility. The advantages of the developed model over the conventional design techniques are also illustrated as it allows for bioretention facilities to be individual assessed based on their specific situations including climate forcing, soil conditions and design parameter sets.

The model allows for long-term simulations to be run in a relatively short amount of time, which would not be possible with more complex equations. It also provides the potential to simulate climate change effects on design given climate series scenarios such as those detailed by the Department for Environment, Food and Rural Affairs (DEFRA) (http://ukclimateprojections.defra.gov.uk/23261).

The model also allows for examination of possible consequences of choosing a design with decreased construction costs such as a low area ratio, which could lead to increased overflow and need for increased maintenance. It can also be used to develop maintenance schedules. This can be achieved by running simulations before constructions whereby an indication of when Pb accumulations reach regulatory limits can be found. These findings can be updated whenever further information is gathered, such as heavy metal runoff concentrations.

In its current state, however the model does not simulate the behavior of other pollutants such as nutrients and hydrocarbons, which are also of concern, but the isotherms for these pollutants can be added to this model in the future.

Despite this, it has been shown that this model can serve as an additional tool available to urban planners and engineers involved in the design, construction and evaluation of SuDS in urban environments.

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Nomenclature

ρ	bulk density of soil
θ	soil moisture content
θ_{res}	residual soil moisture content
θ_{sat}	saturated soil moisture content
A	area ratio of rain garden bioretention system to run on contributing impervious surface
C	dissolved pollutant concentration
C_s	media-sorbed pollutant concentration
D	diffusion coefficient
F	infiltration rate
K_d	linear distribution coefficient
K_{sat}	saturated hydraulic conductivity
R	retardation factor
h	soil–water head
h_s	ponded depth
h_{wf}	average capillary suction at wetting front
t	time
t_p	time of water ponding.
t_r	time at which water input to the soil surface ends
v	pore water velocity
z	distance

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